AI-Driven Digital Evidence Triage in Digital Forensics: A Comprehensive Review

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***Abstract*—Artificial intelligence (AI) has become an essential component in digital evidence triage as a powerful solution that has improved the efficiency of forensic examination. This paper aims to provide a synthesis of recent advancements in the application of AI to digital forensics, based on a review of selected papers from the last decade. It examines the current state-of-the-art methods for using machine learning models, particularly deep learning and hybrid models, in the sorting and examination of evidence. The findings reveal that AI plays a crucial role in addressing challenges such as data complexity, scalability, and anti-forensic countermeasures. However, several challenges remain, including limited datasets, legal issues, and constraints in computational power. This paper emphasizes the importance of a multidisciplinary approach, the emergence of eXplainable Artificial Intelligence (XAI), and the need for a common framework to deal with current and future technologies such as the Internet of Things (IoT), blockchain, and quantum computing in digital forensics. Based on the analysis of current research, this work provides a basis for the development of effective AI-based forensic processes and to cope with the complex nature of contemporary cyber investigations.**

***Index Terms*—Digital Forensics, Artificial Intelligence, Evidence Triage, Machine Learning, Cybercrime**

I. Introduction

Digital evidence triage [1], also referred to as digital forensic triage, is an essential procedure for handling the large quantities of data encountered in contemporary forensic investigations. This technique prioritizes the rapid identification and assessment of relevant evidence throughout the early stages of a case, allowing investigators to allocate resources efficiently and address critical situations. In contrast to traditional forensic methods that require a comprehensive analysis of all digital data, triage emphasizes efficiency and relevance by prioritizing the most critical evidence first.

Conventional forensic approaches usually entail the use of traditional methods such as the collection of data followed by the analysis of the collected data by investigators in order to look for keywords, use filters, and set queries [2]. However, such approaches are inadequate to reveal the hidden patterns or new threats that are not yet visible. Furthermore, the conventional forensic analyses are labor-intensive and time-consuming, especially in fast-paced cases like cybercrime and terrorism investigations. These limitations are compounded by the increasing amount and variety of digital evidence, which calls for innovative and automated approaches.

AI-driven triage leverages machine learning, deep learning, and neural networks to automate pattern recognition, anomaly detection, and digital artifact classification. Traditional methods are static, rule-based;

AI technologies enhance forensic triage through various mechanisms. Natural language processing (NLP) facilitates the automated text analysis that allows investigators to identify important information from emails, chat logs, and social media communications. Computer vision techniques, such as convolutional neural networks (CNNs), rely on deep learning models to identify illicit images and videos. These capabilities enhance efficiency and improve the accuracy of digital evidence assessment, minimizing the risk of overlooking critical information.

This paper examines how AI has transformed digital evidence handling by exploring its applications, challenges, and potential future advancements. Our goal is to provide valuable insights into the use of AI in forensic processes to enhance the accuracy of investigations and optimize the use of available resources and provide an in-depth analysis of current research methods, and propose potential directions for future studies. Figure 1 shows the structure of the paper.

II. Methodology

We adopted a methodical approach [3] to identify, select, and analyze key literature on the integration of artificial intelligence into digital forensics, with an emphasis on selecting the most relevant evidence. Figure 2 illustrates the selection process for this study, narrowing down from 138 papers to 14 papers.

Literature Search Strategy: Multiple academic databases and sites, such as IEEE Xplore, ACM Digital Library, Scopus, ResearchGate, and ConnectedPapers, were searched to ensure the inclusion of all relevant studies.

The search was guided using the following keywords: *"AI in digital forensics," "evidence triage," "machine learning in forensics," "digital forensic triage using machine learning," "Artificial Intelligence on Digital Forensics," and "digital forensic triage using AI."* These keywords were picked to

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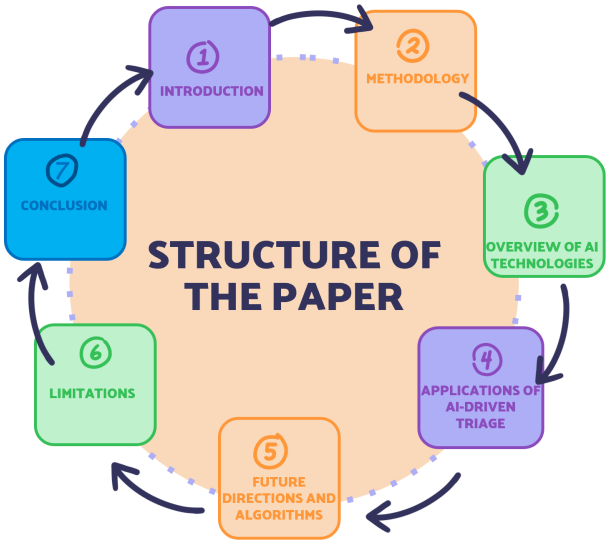


Fig. 1: Paper Structure.

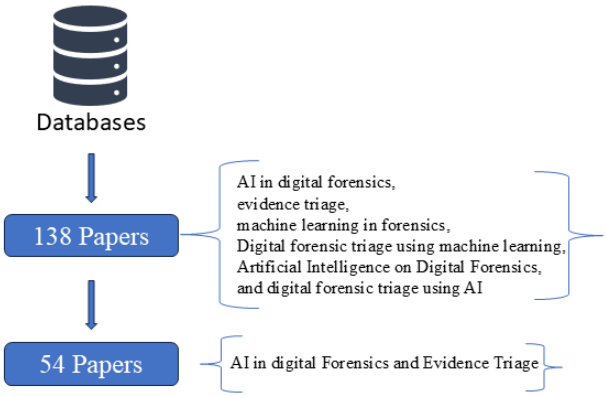


Fig. 2: Paper Selection for this Study.

show how AI and digital evidence triage are related to a lot of different ideas and ways of doing things.

Inclusion and Exclusion Criteria: The inclusion criteria focused on studies published between 2012 and 2024, as the initial timeframe of 2015–2024 yielded a limited number of relevant papers. A total of 138 papers were retrieved through the literature search. Of these, 54 papers were shortlisted based on their relevance to the topic. Further categorization revealed that 40 papers primarily addressed the application of AI in digital forensics but were not directly related to evidence triage. The remaining 14 papers specifically focused on digital forensic triage, and these were selected for detailed analysis. To ensure the quality and reliability of the reviewed studies, we included only peer-reviewed journal articles, book chapters, and conference proceedings.

Analysis Approach: The selected literature was systematically categorized and analyzed based on their methodologies, applications, challenges, and contributions to the field of digital forensic triage. We documented essential aspects of each study, including the use of AI techniques, the context of application, and the outcomes achieved. Thematic areas [4], such as the specific AI algorithms employed (e.g., machine learning, deep learning) and the type of evidence analyzed (e.g., digital images, file systems, multimedia), guided the grouping of studies.

The study focused on the methods and technologies that enhance the accuracy and efficiency of digital forensic triage. It also demonstrated how AI can address longstanding challenges in forensic investigations. The insights gained from this analysis form the basis for the results and discussions presented in the following sections of this review.

III. Overview of AI Technologies in Digital Evidence Triage

The use of artificial intelligence to sort through digital evidence has transformed forensic processes, making them significantly faster, more accurate, and more scalable. New developments, covered in studies, show how machine learning, deep learning, hybrid methods, and specialized frameworks can be used to solve problems in digital forensics.

- **Deep Learning for Image and Video Analysis:** Li and his team [5] designed a system called Human-Centric AI (HXAI) that uses Convolutional Neural Networks (CNNs) to classify images, and it uses Long Short-Term Memory (LSTM) networks to analyze video sequences. These technologies make it possible to quickly identify important evidence, such as firearms or explicit material, within large datasets. According to the HXAI framework, outcomes that are both strong and reliable are guaranteed because it stresses the value of being able to explain results and incorporates human insights. To enhance vulnerability detection, investigation, and mitigation, the HXAI-VDIM [6] system uses explainable AI to integrate human expertise, allowing security analysts to leverage AI recommendations for both proactive and reactive processes.
- **Hybrid Learning Models for Evidence Classification:** For binary and multiclass classification, L. Mohmed et al. [7] created a mixed learning method that combines K-Nearest Neighbors (KNN) and Artificial Neural Networks (ANN). It reduced triage time by 45% while maintaining high accuracy across a wide range of crime categories, such as financial fraud and stalking. This model made considerable steps forward in how digital devices were prioritized.
- **Graph-Based Neural Networks and Blockchain:** Pawar et al. [8] presented a model that integrates a Graph Convolutional Neural Network (GCNN) with blockchain technology to ensure the secure management and classification of digital evidence. Using Natural Language Processing (NLP) techniques has made it easier to analyze text-based evidence in a meaningful way. This shows how flexible AI can be used in forensic investigations.
- **Emerging Forensic Applications in IoT and Smart Grids:** Nelufule et al. [9] examined the forensic chal-

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lenges in smart grids and proposed that AI, IoT, and blockchain technologies could be integrated to improve forensic preparedness. Their study highlights the critical role of AI-driven triage in identifying vulnerabilities in cyber-physical systems and determining which evidence to prioritize during investigations of vital infrastructure.

- **Feature Manipulation for Improved Classification:** McClelland and Marturana [10] created a triage method based on changing features and using both automatic and manual weighting to improve the accuracy of classifying devices. The study used Kullback-Leibler divergence to figure out how important each trait was, showing that it works well in time-sensitive analyses.
- **Machine Learning Approaches:** Supervised machine learning methods, designed for specific types of crimes, have been used in a number of cases. Berte et al. [11] developed a method to categorize digital devices based on features relevant to specific crimes, such as browser information and deleted files. Marturana et al. [12] also demonstrated high classification accuracy by using Bayesian Networks (BN) and Support Vector Machines (SVM) to analyze crimes, such as child abuse and copyright infringement.
- **Interactive Deep Learning Utilizing Human Feedback:** Pirrung et al. [13] introduced Sharkzor, an interactive deep learning tool designed to sort and prioritize large image datasets. The tool uses Few-Shot Learning to adapt to user-selected groups with minimal training examples. By incorporating transparency features, such as confidence visualization, it helps users trust the decisions made by AI systems.
- **Automated Examination for Copyright Violation:** Marturana et al. [14] introduced a machine learning-driven methodology for triage in copyright infringement cases. The study highlighted the use of Bayesian networks and Support Vector Machines (SVM) for evidence classification, achieving accuracies exceeding 93%.
- **Triage for the Series of Photographs and Videos:** Chang et al. [15] introduced a photo triage system utilizing artificial intelligence to rank images in a sequence. This system utilizes pairwise comparisons and Siamese networks to identify high-quality images for investigation, thereby proving to be beneficial for summarizing video keyframes and reviewing extensive media collections.
- **Bulk Data Analysis Utilising Stream-Based Processing:** Garfinkel [16] introduced the *bulk_extractor*, a forensic tool that operates on a stream-based model, enabling the analysis of complete digital media without the necessity for file extraction. Through the application of optimistic decompression and context-sensitive feature extraction, this tool exhibited enhanced efficiency in the identification of significant evidence from extensive datasets.
- **Similarity Digests and Content-Based Triage:** Roussev and Quates [17] presented similarity digests (e.g., sdhash) for content-based triage, facilitating the effective identification of related files across various types of evidence. This methodology enabled swift prioritization of pertinent artifacts, markedly decreasing analysis durations in comparison to conventional techniques.
- **Quantitative Approaches in Mobile Forensics:** Marturana et al. [18] introduced a quantitative triage model for mobile forensics, utilizing machine learning techniques to assess the probability of a device harboring pertinent evidence. This methodology allows researchers to prioritize mobile devices in situations with limited resources.
- **Semi-Automated Triage Frameworks:** Cantrell et al. [19] developed a semi-automated model for the triage process tailored to specific crime scenarios. The proposed framework incorporates live forensics, computer profiling, and keyword searches to enhance the efficiency of identifying high-priority evidence, especially in cases related to child exploitation and hacking.
- **Obstacles in the Implementation of Artificial Intelligence:** Although AI technologies have shown promising results, challenges such as reliance on labeled datasets [11], [12], high computational costs [8], [13], and scalability issues for large datasets continue to present significant obstacles. Current initiatives are focused on mitigating these limitations by employing hybrid methodologies, implementing feature reduction techniques, and developing improved forensic frameworks.

IV. Applications of AI-Driven Triage in Digital Forensics

Building on these advancements, this section follows to further discuss specific applications of AI-driven triage in digital forensics and how these technologies work to address practical challenges and enhance investigative processes. AI-Driven Triage has become an essential component in the field of digital forensics as a solution to problems including the increasing complexity of data, constraints in resources, and the necessity to conduct investigations in a timely manner. These approaches have broad implications across all areas of forensic science, as they are highly effective in lead generation and evidence ranking. This section presents the findings of the study and highlights the gaps that remain unaddressed in the existing research (Table II). The dominant applications include:

- Forensic Triage of Image and Video Evidence: Li and Liu [5] illustrated the application of AI for the rapid identification of essential visual evidence, including weapons, explicit material, and automobiles, within extensive datasets. Forensic analysts can prioritize photos and videos for thorough examination by utilizing Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks.
- Device Prioritization in Resource-Limited Environments: Mohmed et al. [7] suggested a hybrid learning methodology for the prioritization of devices, including smartphones, laptops, and tablets, inside forensic laboratories. Their technique reduced triage time by 45%, allowing

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investigators to focus on devices most likely to harbor relevant evidence.

- Social Media and Textual Forensic Investigations: Pawar et al. [8] emphasized the application of AI in the analysis of text-based evidence on social media platforms. Graph Convolutional Neural Networks (GCNNs) integrated with Natural Language Processing (NLP) methodologies enabled the categorization of data pertaining to cyberbullying, fraud, and several other offences.
- Forensic Preparedness in Smart Grids and IoT Systems: Nelufule et al. [9] investigated the use of artificial intelligence and blockchain technology in smart grids and Internet of Things ecosystems. Their research illustrated how AI-driven triage may improve forensic preparedness by prioritizing evidence collecting and reducing vulnerabilities in principal infrastructure.
- Prioritization of Digital Evidence Specific to Crime: Research by Bertè et al. [11] and Marturana et al. [12] utilized machine learning models to prioritize evidence in instances of child exploitation, hacking, and terrorism. Customized feature extraction facilitated the rapid identification of devices most relevant to specific offenses.
- Triage for Copyright Infringement Cases: An automated triage method for finding evidence of copyright infringement was suggested by Marturana et al. [14]. They employed Bayesian Networks (BN) and Support Vector Machines (SVM) in their study to categorize devices into groups based on factors such as file types and peer-to-peer software traces.
- Mobile Forensics in Urgent Cases: Marturana et al. [18] developed a quantitative priority model for mobile forensics that assists investigators in determining the relative importance of a device to ongoing investigations. Applications include cases of child sexual abuse, homicide, and human trafficking.
- Interactive Instruments for Evidence Organization: Pirrung et al. [13] presented Sharkzor, an interactive deep learning instrument for the categorization and arrangement of evidence. This tool assists investigators in effectively categorizing visually analogous things, facilitating expedited and precise forensic processes.
- Content-Based Triage Utilizing Similarity Digests: Roussev and Quates [17] proposed the use of similarity digests for identifying linked files across various evidence sources. This method facilitated rapid triage in cases of contraband identification and business espionage.
- Analysis of High-Volume Evidence via Bulk Processing: Garfinkel [16] illustrated the utilization of the *bulk_extractor* tool for the analysis of substantial quantities of digital evidence. This technique collected prominent characteristics, including email addresses, URLs, and credit card numbers, significantly decreasing analysis time compared to conventional methods.
- Enhancing Efficiency in Forensic Laboratories: Cantrell et al. [19] suggested a semi-automated triage system that incorporated crime-specific templates, real-time forensics, and keyword searches. This method improved the efficacy of forensic laboratories by optimizing the identification of pertinent evidence. Photo
- Triage for Media Collections: Chang et al. [15] devised an AI-driven method for the evaluation and selection of high-quality images within extensive media libraries. Applications encompass the summarization of video keyframes and the minimization of clutter in photo collections, facilitating the identification of crucial evidence.
- Real-Time Evidence Acquisition and Incident Management: AI-driven triage models have been utilized in real-time contexts, including on-site investigations and incident response. Methods such as optimistic decompression (Garfinkel [16]) and live profiling (Cantrell et al. [19]) facilitate the rapid detection of ephemeral evidence in urgent contexts.

V. Future Directions, Advantages and Disadvantages of Algorithms

There is significant potential for AI-enabled digital evidence triage to revolutionize digital forensics in the future. However, several challenges must be addressed. Table III outlines the challenges and potential solutions for AI-based digital evidence triage. The advantages and disadvantages of the algorithms used in digital evidence triage, including Supervised Learning, Unsupervised Learning, and Semi-Supervised Learning, are outlined in Table I. Several opportunities for further development and improvement are also identified:

- Advancing Real-Time Analysis Capabilities: Future research should focus on optimizing AI models for real-time forensic applications, such as live memory analysis and incident response. Garfinkel [16] emphasized the need for tools capable of handling low-latency scenarios, which can be achieved through advancements in streaming and edge computing technologies.
- Enhancing Dataset Diversity and Quality: The scarcity of labeled forensic datasets remains a significant barrier. Efforts should be directed towards creating large, diverse, and standardised datasets, potentially through collaborations between academic institutions and law enforcement agencies [11] and [12].
- Integrating Explainable AI (XAI): Forensic tools must adopt explainable AI frameworks to ensure transparency and legal admissibility in court proceedings. Li and Liu [5] highlighted the importance of human-centric AI, where interpretability is a core component.
- Addressing Bias in AI Models: Mitigating biases in training datasets and AI decision-making processes is critical. This includes improving feature selection and validation techniques, as discussed by Marturana et al. [10] and [12].
- Developing Forensic Solutions for Emerging Technologies: As IoT devices, smart grids, and blockchain technology become more prevalent, forensic tools must adapt to effectively analyze evidence from these systems. Nelufule

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TABLE I: Advantages and Disadvantages of Algorithms in Digital Evidence Triage

Title	Algorithm Name	Advantages	Disadvantages
Supervised Learning			
Human-Centric AI for Digital Forensic Triage [5]	CNNs	High accuracy in image classification; handles complex features well.	Requires large datasets; computationally intensive.
Human-Centric AI for Digital Forensic Triage [5]	LSTMs	Effective for sequential data to give you an idea video frames.	Complex and computationally expensive for long sequences.
Digital Forensics Triage Classification Model [7]	KNN	Simple, interpretable; effective for smaller datasets.	Struggles with high-dimensional data; sensitive to noise.
Digital Forensics Triage Classification Model [7]	ANN	Good for complex relationships; adaptable to various problems.	Prone to overfitting; requires careful tuning.
GCNN with Blockchain for Digital Forensics [8]	GCNNs	Handles graph data effectively; captures spatial structure.	Computationally expensive; needs high-quality graph inputs.
Crime-Specific Evidence Prioritization [12]	BN	Handles uncertainty well; interpretable.	Requires prior knowledge for probabilities; sensitive to data quality.
Crime-Specific Evidence Prioritization [12]	SVMs	Works well for high-dimensional data; robust to overfitting.	Sensitive to parameter choices; can be slow for large datasets.
Feature Manipulation for Triage [10]	DT	Simple and interpretable; fast on smaller datasets.	Prone to overfitting; less effective on high-dimensional data.
Interactive Deep Learning Tool (Sharkzor) [13]	Few-Shot Learning	Learns with minimal examples; effective for small labeled datasets.	Requires careful tuning; computationally expensive for large datasets.
Unsupervised Learning			
Photo Triage for Media Collections [15]	t-SNE	Good for dimensionality reduction; useful for visualization.	Computationally intensive; does not scale well to large datasets.
Similarity Digests for Content-Based Triage [17]	sdhash	Efficient for detecting similar content; scalable to large datasets.	Limited to content similarity; lacks contextual understanding.
Text-Based Evidence Classification [8]	Topic Modeling	Identifies patterns in unstructured data; adaptable to text analysis.	Depends on high-quality preprocessing; computationally intensive.
Semi-Supervised Learning			
Digital Forensics Triage Classification Model [7]	Hybrid KNN + ANN	Combines strengths of KNN and ANN; effective for classification tasks.	Sensitive to training data quality; computationally expensive.
Photo Triage for Media Collections [15]	Siamese Networks	Learns similarity metrics; adapts with minimal data.	Computationally intensive for large datasets; requires careful training.
GCNN with Blockchain for Digital Forensics [8]	Semi-Supervised GCNNs	Combines graph structure and semi-supervised learning; efficient for sparse labels.	Requires high computational resources; sensitive to graph structure.

et al. [9] emphasized the importance of integrating AI and blockchain for forensic readiness in critical infrastructure.

- Expanding Applications of AI-Driven Triage: AI triage methods should be extended to address emerging cybercrime challenges, such as deepfake detection, ransomware investigations, and advanced persistent threats (APTs). Research into hybrid approaches combining AI with traditional forensic techniques could yield promising results, as evidenced by studies in [8] and [11].

VI. Limitations

This review faced several challenges. First of all, Scope of Research: Digital evidence triage is a sensitive and dynamic domain that necessitates a comprehensive analysis of cutting-edge research, presenting obstacles owing to its intricacy.Secondly, Emerging Nature of the Field: It is challenging to obtain a complete and up-to-date overview of all the evolving changes in digital evidence sorting methods due to their rapid pace of development.

VII. Conclusion

Digital evidence triage using AI has emerged as a significant advancement in digital forensics, offering substantial improvements in speed, accuracy, and scalability. Advanced AI technologies, including deep learning, natural language processing, and hybrid frameworks, enable forensic investigators to prioritize and analyze data more quickly and accurately. Li and Liu [5], Mohmed et al. [7], and Garfinkel [16] have made significant contributions that demonstrate the diverse applications of AI in addressing challenges related to data scale, real-time analysis, and the analysis of data from a wide variety of devices.

Despite these advancements, challenges remain, including limited access to comprehensive datasets, high computational costs, and inherent biases within AI models, which hinder the full realization of AI's potential in digital forensics. Future research should prioritize the development of more transparent and explainable AI systems, the cultivation of specialized forensic skills for emerging technologies, and the fostering of interdisciplinary collaboration among researchers.

In conclusion, AI-driven triage represents a significant advancement in the management of increasingly complex digital evidence. Existing challenges, including bias, efficiency, and user experience, can be addressed through collaborative efforts between researchers and practitioners, focusing on current challenges and exploring promising future directions. This col-

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TABLE II: Contribution and Research Gap Summary

Paper Title	Contribution	Research Gap
Human-Centric AI for Digital Forensic Triage [5]	HXAI framework using CNNs and LSTMs for rapid evidence analysis.	Limited real-time applicability exploration.
Digital Forensics Triage Classification Model [7]	Hybrid KNN and ANN model for binary/multiclass classification.	Needs larger, diverse datasets for generalization.
GCNN with Blockchain for Digital Forensics [8]	GCNN and blockchain for secure evidence and text analysis.	Scalability and computational demands unaddressed.
Smart Grid Forensic Challenges [9]	AI and blockchain for IoT/smart grid forensic readiness.	Lacks implementation details for AI solutions.
Feature Manipulation Techniques for Triage [10]	Feature weighting improves classification accuracy.	Limited testing on diverse forensic datasets.
Machine Learning for Crime-Specific Prioritization [11], [12]	High accuracy for crimes to illustrate exploitation and infringement.	High dependency on labeled datasets, manual features.
Interactive Deep Learning for Evidence Organization [13]	Sharkzor enables Few-Shot Learning-based triage.	Computational efficiency for large datasets challenging.
Automated Triage for Copyright Infringement [14]	Bayesian Networks and SVMs for evidence prioritization.	Scalability for larger datasets unexplored.
Photo Triage for Media Collections [15]	Ranking system using Siamese networks for photo selection.	Limited generalization to forensic use cases.
Bulk Data Analysis with *bulk_extractor* [16]	Stream-based tool for large dataset processing.	Lacks compatibility with emerging IoT data formats.
Similarity Digests for Content-Based Triage [17]	Similarity digests (e.g., sdhash) for evidence identification.	No hybrid AI-driven forensic workflow integration.
Quantitative Triage for Mobile Forensics [18]	Machine learning prioritizes mobile devices in investigations.	Ignores emerging mobile platforms, encryption issues.
Semi-Automated Triage Frameworks [19]	Semi-automated model integrates live forensics and templates.	Needs validation on diverse cases and crimes.

TABLE III: Challenges and Proposed Solutions in AI-Driven Digital Evidence Triage

Challenge	Proposed Solution
Data Dependency and Quality	Large, labeled datasets via collaborations with academia and law enforcement [11], [12].
Scalability for Large Datasets	Stream-based tools suppose that *bulk_extractor* and optimized models for high-volume data [13], [16].
Computational Costs	Hybrid methodologies and efficient model architectures to lower resource demands [8], [13].
Balancing Automation and Human Insights	Explainable AI (XAI) frameworks for integrating human decision-making [5], [19].
Legal Admissibility	Transparent AI models with explainable outputs for courtroom validation [5], [12].
Device Diversity and Compatibility	Standardized tools and methods for diverse devices and formats [9], [18].
Ethical and Privacy Concerns	Privacy-preserving methods and ethical guidelines for evidence collection [8], [19].
Real-Time Data Processing	Low-latency tools and live profiling for time-critical scenarios [16], [19].
Bias in AI Models	Continuous dataset evaluation and improvement to remove biases [5], [11].
Feature Selection and Optimization	Automated and manual feature weighting for crime-specific triage [10], [12].

laborative approach will ultimately contribute to more efficient and effective forensic investigations.

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